

SANDEEP KUMAR KUMAWAT

M.Tech Researcher · Bioscience & Bioengineering · Computational Biology / Bioinformatics

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PROFESSIONAL SUMMARY

M.Tech researcher in Bioscience and Bioengineering at IIT Jodhpur with experience in single-cell RNA-seq, bulk RNA-seq, multi-omics analysis, Gene × Environment (GxE) studies, and microbiological research. Experienced in developing reproducible computational workflows in Python and R for quality control, differential expression, pathway/network analysis, statistical testing, and biological interpretation. Combines computational biology with hands-on microbiology and antimicrobial assay experience.

EDUCATION

M.Tech. (Bioscience & Bioengineering)	Indian Institute of Technology, Jodhpur	CGPA 7.38 (current)	Present
B.Tech. (Biotechnology)	ASPEE SHAKILAM Biotechnology Institute, NAU, Surat	CGPA 8.02	2025

RESEARCH EXPERIENCE

M.Tech Thesis — Single-Cell-Based Approaches for Gene × Environment (GxE) Analysis | IIT Jodhpur | Present

- Investigating cell-type-specific transcriptional responses associated with gene–environment interactions using single-cell transcriptomic approaches.
- Executing end-to-end scRNA-seq workflows covering quality control, preprocessing, clustering, dimensionality reduction, cell-type annotation, and downstream analysis.
- Developing an analytical framework for high-dimensional single-cell data to identify biologically interpretable GxE-associated cellular responses.

Summer Thesis Research — Computational Analysis of Lung Fibrosis | IIT Jodhpur | May 2026 – Jul 2026

- Integrated and batch-corrected three public bulk RNA-seq datasets (GSE110147, GSE24206, GSE99621; 82 samples across Control, Early and Advanced IPF) using ComBat.
- Performed differential-expression analysis and derived an 11-gene common DEG signature; characterized the signature using STRING protein–protein interaction and pathway-enrichment analyses.
- Applied immune-cell deconvolution/correlation analyses and investigated disease-stage molecular patterns relevant to extracellular-matrix remodeling.

Laboratory Intern — MicroCare Laboratory, Pathology Laboratory | Mar 2025 – May 2025

- Performed microbiological and pathological laboratory procedures on 500+ clinical samples, including culture/media preparation and handling of blood, urine, and sputum specimens.
- Performed microscopy, cell counting, antimicrobial activity assays, and antibiotic susceptibility testing across 20 antibiotics.

B.TECH THESIS

Antibacterial Activity of *Murraya koenigii* (Curry Leaf) Leaf Extracts and *Trigonella foenum-graecum* (Fenugreek) Seed Extracts | ASPEE SHAKILAM Biotechnology Institute, NAU, Surat | 2024–2025

- Prepared plant extracts using methanol, acetone, ethyl acetate, and petroleum ether and quantified extractive yield.
- Evaluated antibacterial activity by agar well-diffusion assays against *Bacillus subtilis*, *Staphylococcus aureus*, *Escherichia coli*, *Salmonella typhi*, and *Serratia* across multiple extract concentrations.
- Observed solvent- and concentration-dependent antibacterial activity, with methanolic extracts showing the highest extractive yield for both plant materials.

SELECTED PROJECT

Integrated Omics Analysis of Idiopathic Pulmonary Fibrosis

- Integrated three GEO expression datasets, performed normalization and ComBat batch correction, differential expression, immune-cell fraction analysis, gene–cell fraction correlation, and protein–protein interaction/network analysis.
- Tools included R, Seurat, CIBERSORTx, GSVA, STRING, Cytoscape, limma, and statistical testing.

TECHNICAL SKILLS

Programming: Python, R

Bioinformatics: scRNA-seq, bulk RNA-seq, Seurat, GEO, differential expression, pathway enrichment, CIBERSORTx, GSVA

Data Science: pandas, NumPy, SciPy, Matplotlib, correlation analysis, Wilcoxon test, Kruskal–Wallis test, normalization, ComBat

Tools: Linux/Ubuntu, Jupyter Notebook, RStudio, Git, Cytoscape, STRING

Wet Lab: Microbial culture, solvent extraction, agar diffusion assays, antimicrobial testing, microscopy

PUBLICATION

Book Chapter: "Solvent Extraction-Based Screening of *Murraya koenigii* for Antimicrobial Efficacy against Bacterial Pathogens." Advances in AYUSH, Vol. 17, AkiNik Publications.

RESEARCH INTERESTS

Computational Biology · Bioinformatics · Single-Cell Genomics · Gene × Environment Analysis · Multi-omics · Transcriptomics · Disease Biology · Biomarker Discovery